

MoCA-BG-DET: Motion-Concept Attention Ball-Guided DETR

1 Overview

We propose **MoCA-BG-DET** (Motion-Concept Attention Ball-Guided DETR), a specialized transformer-based detection architecture designed to detect ball-like objects in fixed-camera sports environments.

The model integrates:

- Temporal motion cues extracted from consecutive feature maps
- Concept-guided attention via learned spatial priors
- DETR-based object detection framework

Unlike standard object detectors, the model is explicitly guided toward dynamic, ball-like regions using motion-derived supervision.

2 Input Representation

Let two consecutive frames be:

$$X_t, X_{t-1} \in \mathbb{R}^{B \times 3 \times H \times W} \quad (1)$$

3 Feature Extraction

A shared CNN backbone extracts feature maps:

$$F_t = \text{CNN}(X_t), \quad F_{t-1} = \text{CNN}(X_{t-1}) \quad (2)$$

where:

$$F_t, F_{t-1} \in \mathbb{R}^{B \times C \times H' \times W'} \quad (3)$$

4 Temporal Motion Modeling

Motion is computed using exponentially weighted feature differences:

$$\Delta F = F_t - \gamma F_{t-1} \quad (4)$$

where:

- $\gamma \in [0, 1]$ is a temporal decay factor

The motion energy map is:

$$M_{i,j} = \frac{\|\Delta F_{:,i,j}\|_2}{\max_{i,j} \|\Delta F_{:,i,j}\|_2 + \epsilon} \quad (5)$$

5 Feature Modulation

The feature map is modulated using motion attention:

$$F'_t = F_t \odot (1 + \alpha M) \quad (6)$$

where:

- α controls motion amplification
- $\odot$ denotes element-wise multiplication

6 Transformer Encoding

Flattened features:

$$Z = \text{Flatten}(F'_t) \in \mathbb{R}^{(H'W') \times B \times C} \quad (7)$$

Object queries:

$$Q \in \mathbb{R}^{N_q \times B \times C} \quad (8)$$

7 Concept-Regularized Attention

We define a learned spatial prior $\hat{P}$, trained to approximate motion regions.

The attention mechanism becomes:

$$A = \text{Softmax} \left(\frac{QK^T}{\sqrt{d}} + \lambda \log(\epsilon + \hat{P}) \right) \quad (9)$$

where:

- λ controls prior influence
- ϵ ensures numerical stability

8 Detection Head

Bounding boxes are predicted as:

$$\hat{b} = \sigma(W_b h) \quad (10)$$

where:

- h is the decoder output
- σ is the sigmoid function

9 Loss Function

The total loss combines DETR loss and motion-guided supervision:

$$\mathcal{L} = \mathcal{L}_{DETR} + \beta \cdot \mathcal{L}_{motion} \quad (11)$$

Motion supervision:

$$\mathcal{L}_{motion} = \text{BCE}(\hat{P}, M) \quad (12)$$

where:

- β balances the auxiliary loss

10 Interpretation

The proposed architecture does not rely on color or explicit geometric heuristics. Instead, it learns to focus on ball-like objects by combining:

- Temporal motion consistency
- Attention biasing through learned priors
- Transformer-based global reasoning

11 Conclusion

MoCA-BG-DET provides a structured way to inject motion-based inductive bias into transformer attention, improving detection of small, fast-moving objects such as balls in sports scenarios.